A Subfield Subcode Construction of a Linear Exact Repair Scheme

Nadja Aoutouf and Daniel Augot

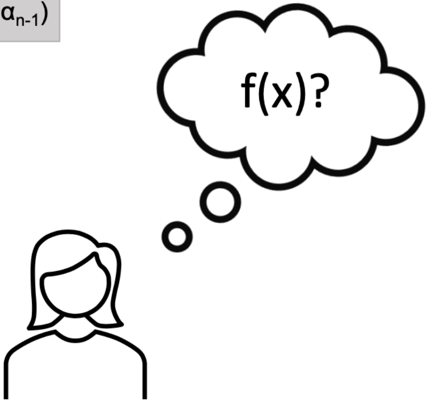
Founded by the Initial Training Network (ITN) ENCODE

The Exact Repair Problem for $[n, k]_{\mathbb{F}_{q^t}}$-RS-Code

$c_0 = f(\alpha_0)$ | $c_1 = f(\alpha_1)$ | $\cdots$ | $c_{n-1} = f(\alpha_{n-1})$

Using Lagrange Interpolation we can reconstruct the polynomial!

$k \cdot t$ $\mathbb{F}_q$-symbols



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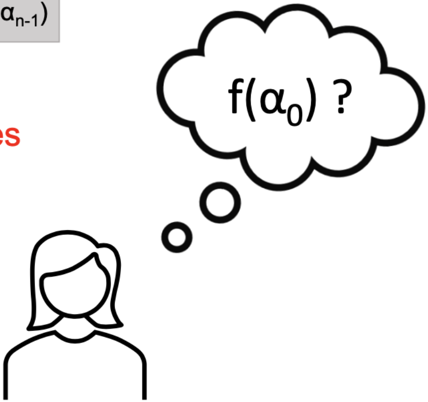

Minimizing the number of nodes contacted
<u>is not the same as</u>
minimizing the amount of F_q-symbols needed!

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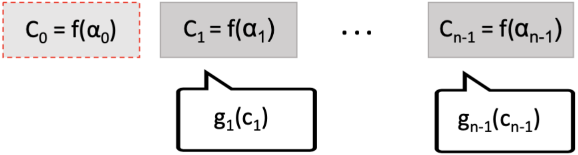
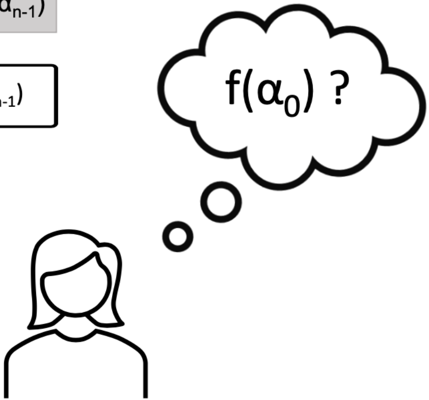
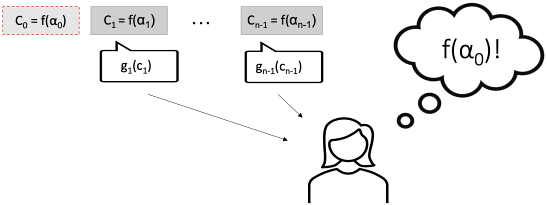


How many $\mathbb{F}_q$-symbols
are needed to reconstruct $f(\alpha_0)$?

Can we do better than the
Lagrange method?



Definition: Linear Exact Repair Scheme



A Linear Exact Repair Scheme (LERS) for an $[n, k]_{\mathbb{F}_{q^t}}$-code C consists for the failed code symbol c_0 of the following

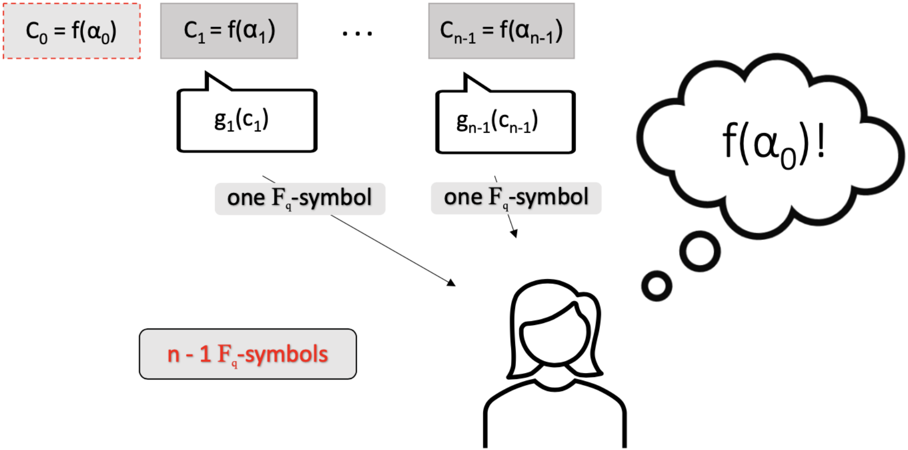
- $\mathbb{F}_q$-linear maps $g_j \colon \mathbb{F}_{q^t} \to \mathbb{F}_q$ for $j \in \{1, \ldots, n-1\}$
- a linear reconstruction map $\mathcal{R} \colon \mathbb{F}_q^{n-1} \to \mathbb{F}_{q^t}$

such that for all $c = (c_0, \ldots, c_{n-1}) \in C$ it holds

$$c_0 = \mathcal{R}(g_1(c_1), \ldots, g_{n-1}(c_{n-1})).$$

Guruswami & Wootters Paper: Main Result

Let $f(X) \in \mathbb{F}_{q^t}[X]$ be a degree $n\left(1 - \frac{1}{q}\right)$ polynomial over $\mathbb{F}_{q^t}$.



Guruswami & Wootters Paper: Main Result[1]

Theorem (Informal)

For $[n, k]_{\mathbb{F}_{q^t}}$-RS-Codes with:

(i) $n = q^t$

(ii) $k \leq n\left(1 - \frac{1}{q}\right)$

a LERS that needs $n-1$ $\mathbb{F}_q$-symbols exist, which is based on t "clever" polynomials $h^{(1)}(X), \ldots, h^{(t)}(X) \in \mathbb{F}_{q^t}[X]_{<q^{t-1}}$, that correspond to t dual codewords.

That is one $\mathbb{F}_q$-symbol per surviving node

$$n - 1 = n\left(1 - \frac{1}{n}\right)$$

The Lagrange method needs for largest k

$$kt = n\left(1 - \frac{1}{q}\right)t$$

[1] V. Guruswami and M. Wootters: *Repairing Reed–Solomon Codes*, STOC 2016.

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Definitions

A linear code C is **non-degenerate** at position 0 if there exists a codeword in C with a nonzero coordinate at position 0.

For an $[n, k]$-code C, the **punctured code** at position 0, denoted by $\underline{C}$, is the code of length $n-1$ obtained by deleting the code symbol on position 0.

Given $(-1, \phi_1, \ldots, \phi_{n-1}) \in C$, an **extension map** for the dual of a non-degenerate code C is defined as

$$\phi_{\underline{(C^\perp)}} : \begin{cases} \underline{(C^\perp)} \longrightarrow \mathbb{F}_{q^t} \\ \underline{h} \longmapsto \sum_{j=1}^{n-1} \phi_j h_j = h_0. \end{cases}$$

Note that $(\phi_{\underline{(C^\perp)}}(\underline{h}), \underline{h}) \in C^\perp$.

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Theorem

Let $C \subseteq \mathbb{F}_{q^t}$ be a non-degenerate linear code and let c_0 be a failed code symbol. Then a LERS that needs at most $n-1$ $\mathbb{F}_q$-symbols can be constructed from t dual codewords $h^{(1)}, \ldots, h^{(t)} \in C^{\perp}$ satisfying

(i) $\dim_{\mathbb{F}_q} \langle h_0^{(1)}, \ldots, h_0^{(t)} \rangle = t$

(ii) $\dim_{\mathbb{F}_q} \langle h_j^{(1)}, \ldots, h_j^{(t)} \rangle \leq 1 \qquad$ *for all* $j \in \{1, \ldots, n-1\}$.

$\rightarrow$ That is one sub-symbol per surviving node!

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Problem Statement in Matrix Form

Find a LERS for a non-degenerated code $C \subseteq \mathbb{F}_{q^t}$ that uses at most $n-1$ $\mathbb{F}_q$-symbols.

$$\Updownarrow$$

Find H_t that consists out of $h^{(1)}, \ldots, h^{(t)} \in C^\perp$:

$$H_t = \begin{bmatrix} h^{(1)} \\ h^{(2)} \\ \vdots \\ h^{(t)} \end{bmatrix} = \begin{bmatrix} h_0^{(1)} & h_1^{(1)} & \cdots & h_{n-1}^{(1)} \\ h_0^{(2)} & h_1^{(2)} & \cdots & h_{n-1}^{(2)} \\ \vdots & \vdots & \ddots & \vdots \\ h_0^{(t)} & h_1^{(t)} & \cdots & h_{n-1}^{(t)} \end{bmatrix}$$

$$\text{s.t.} \quad \dim_{\mathbb{F}_q} \langle h_0^{(1)}, \ldots, h_0^{(t)} \rangle = t \quad \text{and} \quad \dim_{\mathbb{F}_q} \langle h_j^{(1)}, \ldots, h_j^{(t)} \rangle \leq 1 \quad \forall j \in \{1, \ldots, n-1\}$$

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Main Contribution

Experimental Result

For every non-degenerate $[n, k]$-code over $\mathbb{F}_{q^t}$, there exists, with high probability, a linear exact repair scheme that uses at most $n-1$ $\mathbb{F}_q$-symbols whenever

$$k \leq \left\lfloor \frac{n-1}{t} \right\rfloor - 1.$$

Subfield Subcode Construction of an LERS

Goal

Given a non-degenerated $[n,k]_{\mathbb{F}_{q^t}}$-code C with $k \leq \left\lfloor \frac{n-1}{t} \right\rfloor - 1$. Find $\beta_1, \dots, \beta_{n-1} \in \mathbb{F}_{q^t}$ in order to construct $h^{(1)}, \dots, h^{(t)} \in C^{\perp}$ such that:

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(ii) $\langle h_j^{(1)}, \dots, h_j^{(t)} \rangle_{\mathbb{F}_q} = \langle \beta_j \rangle_{\mathbb{F}_q}$ for all $j \in \{1, \dots, n-1\}$

Sketch of construction: Rewrite any dual codeword $h \in C^{\perp}$ as $h = (h_0, \underline{h})$.

1. Construction of an non-empty subcode $\underline{(C_{\beta}^{\perp})}$ that contains all $\underline{h} \in \underline{(C^{\perp})}$ such that condition (ii) is fulfilled.
2. The $\underline{(C_{\beta}^{\perp})}$-extension map sends $\underline{h} \in \underline{(C_{\beta}^{\perp})}$ to $h_0 \in \mathbb{F}_{q^t}$ such that $(h_0, \underline{h}) \in C^{\perp}$.
3. Find $\beta_1, \dots, \beta_{n-1} \in \mathbb{F}_{q^t} \setminus \{0\}$ such that there exist $\underline{h}^{(1)}, \dots, \underline{h}^{(t)} \in \underline{(C_{\beta}^{\perp})}$ that map on $\mathbb{F}_q$-linearly independent $h_0^{(1)}, \dots, h_0^{(t)}$.

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Define

$$\underline{(C^{\perp}_{\beta})} := \{\underline{h} \in \underline{(C^{\perp})} \mid h_j \in \langle \beta_j \rangle_{\mathbb{F}_q} \text{ for all } j\} \subseteq \mathbb{F}_{q^t}^{n-1}.$$

- Since $h_j = \lambda_j \beta_j$ for some $\lambda_j \in \mathbb{F}_q$, define

$$\underline{\Lambda_{\beta}} := \{(\lambda_1, \ldots, \lambda_{n-1}) \in \mathbb{F}_q^{n-1} \mid \exists\ \underline{h} \in \underline{(C^{\perp}_{\beta})} \text{ s.t. } h_j = \lambda_j \beta_j \text{ for all } j\}.$$

- Note that $\underline{\Lambda_{\beta}}$ is a subfield subcode:

$$\underline{\Lambda_{\beta}} = \underline{(C^{\perp}_{\beta})} \cdot Diag(\beta_1^{-1}, \ldots, \beta_{n-1}^{-1}) \cap \mathbb{F}_q^{n-1}.$$

- Delsarte's theorem: $\dim_{\mathbb{F}_q}(\underline{\Lambda_{\beta}}) \geq (n-1) - kt$.
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$$\underline{(C_\beta^\perp)} := \{\underline{h} \in \underline{(C^\perp)} \mid h_j \in \langle \beta_j \rangle_{\mathbb{F}_q} \text{ for all } j\} \subseteq \mathbb{F}_{q^t}^{n-1}.$$

- Since $h_j = \lambda_j \beta_j$ for some $\lambda_j \in \mathbb{F}_q$, define

$$\underline{\Lambda_\beta} := \{(\lambda_1, \ldots, \lambda_{n-1}) \in \mathbb{F}_q^{n-1} \mid \exists\ \underline{h} \in \underline{(C_\beta^\perp)} \text{ s.t. } h_j = \lambda_j \beta_j \text{ for all } j\}.$$

- Note that $\underline{\Lambda_\beta}$ is a subfield subcode:

$$\underline{\Lambda_\beta} = \underline{(C_\beta^\perp)} \cdot Diag(\beta_1^{-1}, \ldots, \beta_{n-1}^{-1}) \cap \mathbb{F}_q^{n-1}.$$

- Delsarte's theorem: $\dim_{\mathbb{F}_q}(\underline{\Lambda_\beta}) \geq (n-1) - kt$.
- $k < \frac{n-1}{t} \implies \dim_{\mathbb{F}_q}(\underline{\Lambda_\beta}) > 0 \implies \underline{(C_\beta^\perp)}$ is non-empty.

Step 2: Given $\underline{h} \in \underline{(C_\beta^\perp)}$ find $h_0 \in \mathbb{F}_{q^t}$ such that $(h_0, \underline{h}) \in C^\perp$

- There exists $(-1, \phi_1, \ldots, \phi_{n-1}) \in C$.
- Define the extension map as

$$\phi_{\underline{(C_\beta^\perp)}} : \begin{cases} \underline{(C_\beta^\perp)} \longrightarrow \mathbb{F}_{q^t} \\ \underline{h} \longmapsto \sum_{j=1}^{n-1} \phi_j h_j := h_0 \end{cases}$$

- Hence $(\phi_{\underline{(C_\beta^\perp)}}(\underline{h}), \underline{h}) \in C^\perp$.

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Step 3: Find $\beta_1, \ldots, \beta_{n-1} \in \mathbb{F}_{q^t} \backslash \{0\}$ s.t. $\dim_{\mathbb{F}_q} \langle h_0^{(1)}, \ldots, h_0^{(t)} \rangle = t$

It is enough to show that $\dim_{\mathbb{F}_q}(\mathrm{Im}(\phi_{\underline{(C_\beta^\perp)}})) = t$.

- Define

$$\psi_{\underline{\Lambda_\beta}} : \begin{cases} \underline{\Lambda_\beta} \longrightarrow \mathbb{F}_{q^t} \\ (\lambda_1, \ldots, \lambda_{n-1}) \longmapsto \sum_{j=1}^{n-1} \phi_j \lambda_j \beta_j := h_0 \end{cases}$$

- $\dim_{\mathbb{F}_q}(\mathrm{Im}(\phi_{\underline{\Lambda_\beta}})) = t \iff \dim_{\mathbb{F}_q}(\mathrm{Im}(\psi_{\underline{\Lambda_\beta}})) = t \implies \dim_{\mathbb{F}_q}(\underline{\Lambda_\beta}) \geq t.$

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We know:

$$\dim_{\mathbb{F}_q}(\mathrm{Im}(\phi_{\underline{\Lambda_\beta}})) = t \implies \dim_{\mathbb{F}_q}(\underline{\Lambda_\beta}) \geq t.$$

- By Delsarte's theorem, $\dim_{\mathbb{F}_q}(\underline{\Lambda_\beta}) \geq (n-1) - kt$.
- Using Delsarte's Theorem, a sufficient condition for $\dim_{\mathbb{F}_q}(\underline{\Lambda_\beta}) \geq t$ is:

$$(n-1) - kt \geq t \quad \iff \quad k \leq \frac{n-1}{t} - 1.$$

Since $\dim_{\mathbb{F}_q}(\mathrm{Im}(\phi_{\underline{\Lambda_\beta}}))$ is hard to compute explicitly, we use a **randomized method** to find the $\beta_1, \ldots, \beta_{n-1} \in \mathbb{F}_{q^t}$. Here, the derived threshold for k increases the success probability.

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Randomized Method

Input: Parameters q, t, n and k.

Output: `True` if $\beta_1, \ldots, \beta_{n-1}$ are found in one trial, `False` otherwise.

1. Sample $\beta_1, \ldots, \beta_{n-1} \overset{\$}{\leftarrow} \mathbb{F}_{q^t} \backslash \{0\}$.
2. Check: $\dim_{\mathbb{F}_q} \left(\mathrm{Im}(\phi_{\underline{\Lambda_\beta}}) \right) = t$?
   - ✓ Yes $\Rightarrow$ Return: `True`
   - × No $\Rightarrow$ Return: `False`

For $k \leq \frac{n-1}{t} - 1$, the randomized method succeeds with high probability.

$\Longrightarrow$ **We can find suitable $\beta_1, \ldots, \beta_{n-1}$ with high probability.**

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Experimental Results of the Randomized Method

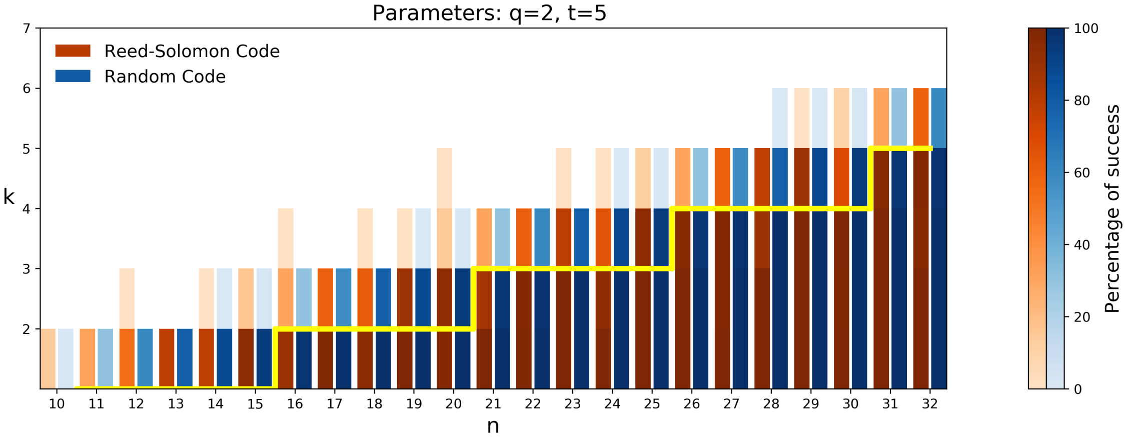

Figure: Percentage of success (for 10000 trials for each pair (n, k)) for $q = 2$ and $t = 5$. The threshold for k is indicated in yellow.

Comparison with the GW Linear Repair Scheme

GW Linear Repair Scheme

- Explicit for **Reed-Solomon codes**.

Parameters Allowed

$$n = q^t, \qquad k \leq n\left(1 - \frac{1}{q}\right)$$

Advantage: Less restrictive parameter constraints

Limitation: More restrictive in code structure.

Our Linear Repair Scheme

- Applicable to **any non-degenerated code**.

Parameters Allowed

$$k \leq \left\lfloor \frac{n-1}{t} \right\rfloor - 1$$

Advantage: No strong algebraic structure needed.

Limitation: Slightly weaker parameters than GW scheme.

Questions?

Proof Idea

Lets derive the scheme for $k = \frac{n}{2}$, $n = 2^t$ and $\alpha * = 0$.

- It can be shown that $\forall f, h \in \mathbb{F}_{2^t}[X]$ with degree smaller than $\frac{n}{2}$

$$0 = \sum_{j \in \{1,\dots,n\}} f(\alpha_j) \cdot h(\alpha_j)$$

- Choose h "clever": Fix any basis $Z = \{\zeta_1, \cdots, \zeta_t\}$ of $\mathbb{F}_{2^t}$ over $\mathbb{F}_2$. Pick any $\zeta \in Z$ and set

$$h(X) = \frac{\mathrm{Tr}(\zeta \cdot X)}{X} = \zeta + \zeta^2 X + \zeta^4 X^3 + \cdots + \zeta^{\frac{n}{2}} X^{\frac{n}{2}-1}$$

Facts about h:

$$deg(h) < \frac{n}{2} \qquad h(0) = \zeta \qquad h(\alpha_j) = \frac{\mathrm{Tr}(\zeta \cdot \alpha_j)}{\alpha_j}$$

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This results in the following:

$$0 = f(0) \cdot h(0) + \sum_{\alpha_j \neq 0} f(\alpha_j) \cdot h(\alpha_j)$$

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Proof Idea

Proposition

Given a fixed basis $Z = \{\zeta_1, \cdots, \zeta_t\}$ of $\mathbb{F}_{2^t}$ over $\mathbb{F}_2$, then each symbol $f(\alpha*) \in \mathbb{F}_{2^t}$ can be recovered from its t independent traces $\mathrm{Tr}(\zeta_i f(\alpha*))$ using the dual basis of Z (denoted by $V = \{\nu_1, \cdots, \nu_t\}$):

$$f(\alpha*) = \sum_{i \in \{1,\ldots,t\}} \mathrm{Tr}(\zeta_i f(\alpha*))\nu_i$$

Proof Idea

Proof.

As $f(\alpha*) \in \mathbb{F}_{2^t}$ it can be written as linear combination:

$$f(\alpha*) = \sum_{i \in \{1,\dots,t\}} b_i \nu_i \qquad \text{for } b_i \in \mathbb{F}_2 \tag{1}$$

Note that

$$\begin{aligned} \mathrm{Tr}(\zeta_j f(\alpha*)) &= \mathrm{Tr}(\zeta_j \sum_{i \in \{1,\dots,t\}} b_i \nu_i) \\ &= \mathrm{Tr}(\sum_{i \in \{1,\dots,t\}} b_i \zeta_j \nu_i) \\ &= \sum_{i \in \{1,\dots,t\}} b_i \,\mathrm{Tr}(\zeta_j \nu_i) \\ &= b_j \end{aligned}$$

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